FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2019

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

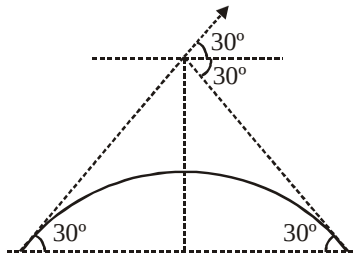
Physics

PART – I

SECTION – A

1. BC

Sol. $T = \dfrac{2u\sin\theta}{g} = \dfrac{2\times 10\times\frac{1}{2}}{10} = 1s.$



2. ACD

Sol. As $\dfrac{1}{\lambda} = R\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right)$ and $P = \dfrac{h}{\lambda}$

Also, $E = \dfrac{hc}{\lambda}$

3. AD

Sol. $\because\ \vec{B} \perp (\vec{A}\times\vec{B})$

$\vec{A}.\left(\vec{C}+\vec{D}\right) = 0$, $\vec{A}.\vec{C} + \vec{A}.\vec{D} = 0$

Component of $\vec{C}$ along $\vec{A}$ =– component of $\vec{D}$ along $\vec{A}$

4. BC

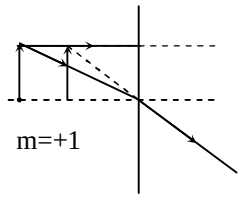
Sol. $|f - 256| = 2Hz$

$f = (256 \pm 2)Hz = (254)Hz$ or $(258)Hz$

If tension increases, then f′ may be 259 Hz.

5. AC

Sol. Apparent position of the object wrt lens.



$$u = \left(\frac{10}{1} + \frac{10}{2}\right) = 15\,cm$$

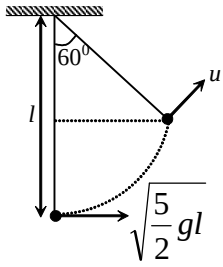
$$\frac{1}{15} + \frac{1}{v} = \frac{2(1.5) - 1}{R}$$

v = 7.5 cm

Image size = $\frac{7.5}{15} \times 1$ = 0.5 cm

6. ABCD

Sol. $T - mg\cos 60^0 = \frac{mv^2}{l}$ … (i)



$$\frac{1}{2}m\left(\sqrt{\frac{5}{2}gl}\right)^2 - mg\frac{l}{2} = \frac{1}{2}mv^2 \quad \text{… (ii)}$$

$$v = \sqrt{\frac{3}{2}gl} \text{ and } T = 2mg$$

Net force is in horizontal direction when $\theta = 60^0$

7. C

Sol. $P = 5P_o - \frac{P_o V}{V_o}$

$$T = \frac{1}{nR}\left(5P_o V - \frac{P_o V^2}{V_o}\right)$$

$$\frac{dT}{dV} = \frac{1}{nR}\left(5P_o - \frac{2P_o V}{V_o}\right) = 0$$

$\Rightarrow V = \frac{5V_o}{2}$ at maximum temperature.

$\Rightarrow T_{max} = \frac{25}{16}T_o$

8. D

Sol. $\frac{dQ}{ndT} = C_v + \frac{P}{n}\frac{dV}{dT} = 0$

$$C_V + \frac{\left(5P_o - \frac{P_o V}{V_o}\right)}{n} \cdot \frac{nR}{\left(5P_o - \frac{2P_o V}{V_o}\right)} = 0$$

$$\frac{\left(5-\frac{V}{V_o}\right)}{\left(5-\frac{2V}{V_o}\right)}=-\frac{3}{2} \qquad \Rightarrow V=\frac{25}{8}V_o$$

9. D

Sol. $E=\frac{\rho}{4\varepsilon_o}r$

$V_{AB}=E\times\frac{r}{2}$

10. A

Sol. V_{CD} = Er cos 90° = 0

SECTION – D

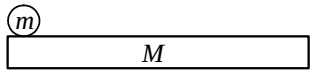
11. 00002.00

Sol. $X_{cm}=\frac{\ell}{2\sqrt{2}}, I=2m\frac{\ell^2}{3}$

$$T=2\pi\sqrt{\frac{I}{2mgcm}}=2\pi\sqrt{\frac{2\sqrt{2}\ell}{3g}}$$

12. 00001.40

Sol. $V_{mM}=4\hat{i}+3\hat{j}$



$V_m=u\hat{i}+3\hat{j}$

$V_M=-V\hat{i}$

Here, along horizontal direction $mV_m+MV_M=0$

$\Rightarrow \quad V=\frac{um}{M}=u\eta$ and $u+V=4 \Rightarrow u=\frac{4}{1+\eta}$

To move one meter w.r.t. ground

$$1=\frac{2\times\frac{4}{1+\eta}\times 3}{g} \Rightarrow 1+\eta=\frac{24}{g} \text{ or } \eta=\frac{14}{10}=1.40$$

13. 00000.75

Sol. $(Fs)_{max}$ = 20 N. a_{max} of 20 kg black = 1 m/s^2

$\therefore$ both black move together till 6 sec.

$\therefore \quad \frac{dv}{dt}=\frac{5t}{30} \quad \Rightarrow \int dv=\frac{1}{6}\ \frac{t^2}{2}$; $V=\frac{3^2}{12}=\frac{9}{12}$ = 0.75 m/s

14. 00000.39

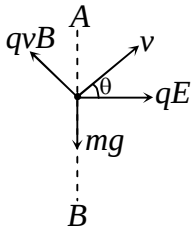
Sol. Applying Snell's law between the points *O* and *P*, we have

$$2\times\sin 60^0=\left(\sin 90^0\right)\times\frac{2}{\left(1+H^2\right)}, \quad 2\times\frac{\sqrt{3}}{2}=1\times\frac{2}{\left(1+H^2\right)}$$

$$\left(1+H^2\right)=\frac{2}{\sqrt{3}},\quad H=\sqrt{\left(\frac{2}{\sqrt{3}}-1\right)}$$

15. 00019.60

Sol. When crosses *AB*



$$qvB_0\cos\theta=mg$$

$$qvB_0\sin\theta = qE$$

$$\tan\theta=\frac{qE}{mg}=1$$

$$\theta=\frac{\pi}{4}$$

along horizontal $v\cos\theta=\frac{qE}{m}t_0$

$$u-gt_0=v\sin\theta$$

$$u=\left(g+\frac{qE}{m}\right)t_0 \;= 2gt_0$$

16. 00000.23

Sol. $\mu=\dfrac{\text{velocity in air}}{\text{velocity in medium}}$

17. 00005.00

Sol. $W=\frac{D\lambda}{d}, W'=\frac{D}{d}\left(\lambda-\frac{\lambda}{5}\right) \quad \Rightarrow W'=\frac{D}{d}\frac{\lambda}{5/4}$

18. 00002.00

Sol. $\frac{q_1}{C_1}=\frac{q_2}{C_2}$; $q_1+q_2=2Q_0$

$$C_1=\frac{\varepsilon_0 A}{d_0+vt};\ C_2=\frac{\varepsilon_0 A}{d_0-vt}$$

$$\frac{q_1}{q_2}=\frac{d_0-vt}{d_0+vt}$$

$$q_2\left(\frac{d_0-vt}{d_0+vt}\right)+q_2=2Q_0$$

$$q_2\left[\frac{2d_0}{d_0+vt}\right]=2Q_0$$

$$q_2=\frac{2Q_0}{2d_0}\left(d_0+vt\right)$$

$$I=\frac{dq_2}{dt}=\frac{Q_0 v}{d_0}=2 \text{ amp}$$

Chemistry

PART – II

SECTION – A

19. BC

Sol. V_{CH_3OH} = 30 mL

d = 0.8 g/cc

$W = 0.8 \times 30 = 24$ g

$n = \frac{24}{32} = 0.75$

$V_{C_2H_5OH}$ = 60 mL

d = 0.92 g/cc

$W = 60 \times 0.92 = 55.2$ g

$n = \frac{55.2}{46} = 1.2$

Total moles = 0.75 + 1.2 = 1.95

Solute = CH_3OH, Solvent = C_2H_5OH

Molarity = $0.75 \times \frac{1000}{90} = 8.33$

Molality = $0.75 \times \frac{1000}{55.2} = 13.59$

Mole fraction of solute (CH_3OH) = $\frac{0.75}{1.2+0.75} = 0.385$

Mole fraction of solvent (C_2H_5OH) = 1 - 0.385 = 0.615

20. ABC

Sol. $Na_3\left[Co(ONO)_6\right] \xrightarrow{H_2O} 3Na^+ + \left[Co(ONO)_6\right]^{3-}$

$\left[Co(ONO)_6\right]^{3-} + H_2O \longrightarrow H\left[Co(ONO)_6\right]^{2-} + OH^-$

The IUPAC name of the ligand is nitrito.

21. ABC

Sol. For option (A)

$P = 80X_A + 240\ X_B$

$= 80\ X_A + 240\ (1 - X_A)$

$= 80\ X_A + 240 - 240\ X_A$

$= 240 + X_A(80 - 240)$

$= 240 – 160\ X_A$

$\therefore$ V.P of solution is less than 240 mm Hg which is V.P of pure(B)

For option (B)

$P = 80X_A + 240\ X_B$

$= 80(1 – X_B) + 240\ X_B$

$= 80 – 80\ X_B + 240\ X_B$

$= 80 + (240 – 80)X_B = 80 + 160\ X_B$

$\therefore$ V.P of solution is more than 80 mm Hg which is the V.P of pure(A)

For option (C)

$P = 80X_A + 240X_B$

For pure (A), $X_A = 1$ and $X_B = 0$

V.P of pure (A) = 80 mm Hg

Similarly

$\therefore$ V.P of pure (B) = 240 mm Hg.

22. CD

Sol. $Co(BO_2)_2$ and $Cu(BO_2)_2$ are blue substances.

23. BD

Sol. Metals having higher reduction potentials are reduced in presence metals having lower reduction potentials.

24. ABC

Sol. If a solution is prepared with a fixed pH or molarity, it maintains its characteristics whatever amount of it, are used.
Refractive index is fixed for any substance and is mass independent.

25. B

Sol. P = $Cr_2(SO_4)_3$, Q = Na_2CrO_4, R = $Na_2Cr_2O_7$, S = $PbCrO_4$

26. C

Sol. P = $Cr_2(SO_4)_3$, Q = Na_2CrO_4, R = $Na_2Cr_2O_7$, S = $PbCrO_4$

27. B

Sol. At cathode: $2H_2O + 2e \longrightarrow H_2 + 2OH^-$

At anode: $2Cl^- \longrightarrow Cl_2 + 2e$

Among the given electrolytes only sodium is present above hydrogen in the activity series. Hence aqueous NaCl should be electrolysed. Other metals like Cu, Hg and Au are present below hydrogen in the activity series.

28. C

Sol. At cathode: $2H_2O + 2e \longrightarrow H_2 + 2OH^-$

At anode: $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

Since K and Al are present above hydrogen in the activity series, H_2 gas will be liberated at the cathode and O_2 at the anode.

SECTION – D

29. 00143.50

Sol. The complex is $[Cr(NH_3)_4Cl_2]Cl$
$\therefore$ Mole of AgCl formed = 1
Mass of AgCl = 143.50 g

30. 00006.38

Sol. $pH = pK_a + \log\frac{[NaA]}{[HA]}$

or, $5.38 = pK_a + \log\frac{1}{10}$

$\therefore$ $pK_a = 5.38 + 1 = 6.38$

31. 02078.50

Sol. $\Delta H = nC_P\Delta T$

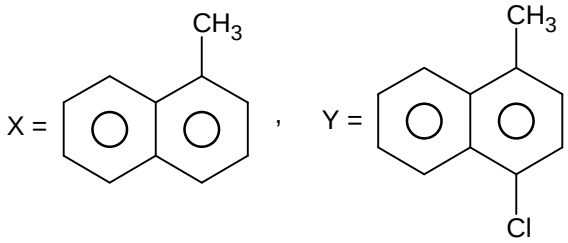
$= \frac{4}{4} \times \frac{5}{2} \times 8.314 \times (140 - 40) = 2078.5$ J

32. 00073.50

Sol. P = $CHCl_3$(Molar mass = 119.5)
Q = HCOOH(molar mass = 46)
$\therefore$ P – Q = 119.5 – 46 = 73.5

33. 00176.50

Sol.



34. 00030.11

Sol. $d = \frac{Z \times A}{N \times a^3} \Rightarrow 1.5 = \frac{0.03 \times A}{6.022 \times 10^{23} \times \left(10^{-8}\right)^3}$

On solving A = 30.11

35. 01690.00

Sol. $\Delta T_f = K_f \times m = K_f \times \frac{w}{M} \times \frac{1000}{W}$

or, $0 - (-6) = 1.86 \times 1 \times \frac{1000}{W}$

On solving W = 310 g
$\therefore$ Mass of ice = 2000 – 310 = 1690.00 g

36. 00082.10

Sol. $P = \frac{nRT}{V} = \frac{100 \times 0.0821 \times 1000}{100} = 82.1 \text{ atm}$

Mathematics

PART – III

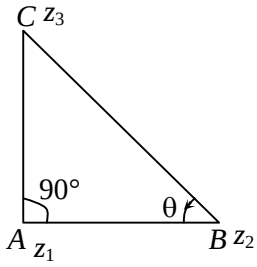
SECTION – A

37. AB

Sol. $\frac{W_2}{|W_2|} = \frac{W_1}{|W_1|} e^{-i\theta}$

$\Rightarrow \quad \frac{W_2}{W_1} = \frac{|W_2|}{|W_1|} e^{-i\theta} = 1 - i\tan\theta$

$W_1\overline{W}_2 = \frac{W_1}{W_2}|W_2|^2 = |W_1|\ |W_2| e^{i\theta} = \left(|W_1|^2 + i|W_1|\ |W_2|\tan\theta\right)$



38. ABC

Sol. We have $A(\alpha,\beta)' = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & e^{\beta} \end{bmatrix}$

$= \begin{bmatrix} \cos(-\alpha) & \sin(-\alpha) & 0 \\ -\sin(-\alpha) & \cos(-\alpha) & 0 \\ 0 & 0 & e^{\beta} \end{bmatrix} = A(-\alpha, \beta)$

Also, $A(\alpha,\beta)A(-\alpha,-\beta)$

$= \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & e^{\beta} \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & e^{-\beta} \end{bmatrix}$

$= I$

$\Rightarrow A(\alpha,\beta)^{-1} = A(-\alpha, -\beta)$

Next, $\text{Adj}(\alpha,\beta) = |A(\alpha,\beta)| A(\alpha,\beta)^{-1}$

$= e^{\beta} A(-\alpha, -\beta)$

39. AB

Sol. $I = \int_0^{\alpha} \frac{dx}{1 - \cos\alpha\cos x}$

$= \int_0^{\alpha} \frac{dx}{\left(\cos^2\frac{x}{2} + \sin^2\frac{x}{2}\right) - \cos\alpha\left(\cos^2\frac{x}{2} - \sin^2\frac{x}{2}\right)}$

$= \int_0^{\alpha} \frac{dx}{(1-\cos\alpha)\cos^2\frac{x}{2} + (1+\cos\alpha)\sin^2\frac{x}{2}}$

$= \int_0^{\alpha} \frac{dx}{2\sin^2\frac{\alpha}{2}\cos^2\frac{x}{2} + 2\cos^2\frac{\alpha}{2}\sin^2\frac{x}{2}}$

$$= \frac{1}{2}\int_0^{\alpha} \frac{\left(\sec^2\frac{\alpha}{2}\right)\sec^2\frac{x}{2}}{\tan^2\frac{\alpha}{2}+\tan^2\frac{x}{2}}dx$$

Put $\tan\frac{x}{2} = t$, we have

$$I = \int_0^{\tan\frac{\alpha}{2}} \frac{\sec^2\frac{\alpha}{2}\,dt}{t^2+\tan^2\frac{\alpha}{2}}$$

$$= \sec^2\frac{\alpha}{2}\cot\frac{\alpha}{2}\left[\tan^{-1}\left(\frac{t}{\tan\frac{\alpha}{2}}\right)\right]_0^{\tan\frac{\alpha}{2}}$$

$$= \frac{2}{\sin\alpha}.\frac{\pi}{4} = \frac{\pi}{2\sin\alpha}$$

Thus $\frac{A}{\sin\alpha}+B = \frac{\pi}{2\sin\alpha}.A = \frac{\pi}{2}$, $B = 0$ and $A = \frac{\pi}{4}$,

$B = \frac{\pi}{4\sin\alpha}$ satisfy the last equation.

40. A

Sol. Integrating the given differential equation, we have $\frac{dy}{dx} = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + C_1$ but

$y_1(0) = 1$ so $1 = \left(\frac{-1}{3}\right) + 1 + C_1 \Rightarrow C_1 = \frac{1}{3}$.

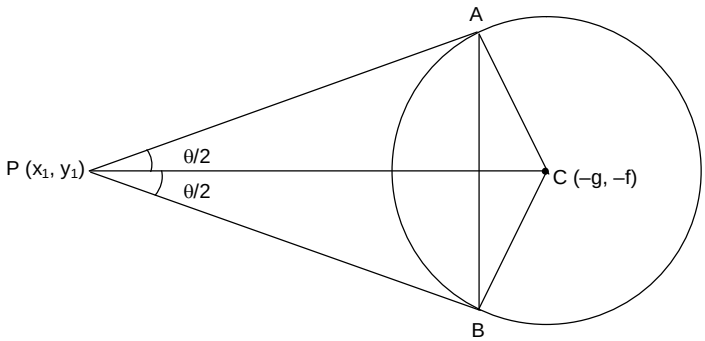
Again integrating, we get $y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x + C_2$

but $y(0) = 0$ so $0 = 0 + 1 + C_2 \Rightarrow C_2 = -1$. Thus $y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$

41. BD

Sol. Let PA and PB be the tangents from $P(x_1, y_1)$ to the given circle with centre $C(-g-f)$, such that $\angle APB = \theta$. Then $\angle APC = \frac{\theta}{2}$ (Figure). Therefore, from ΔPAC, we get

$$\cot\frac{\theta}{2} = \frac{PA}{AC} = \frac{\sqrt{S_1}}{\sqrt{g^2+f^2-c}}$$



$$\Rightarrow \tan\frac{\theta}{2}=\frac{\sqrt{g^2+f^2-c}}{\sqrt{S_1}}$$

$$\Rightarrow \theta=2\tan^{-1}\left(\frac{\sqrt{g^2+f^2-c}}{\sqrt{S_1}}\right)$$

42. BD

Sol. $\sin A+\sin B=\frac{a+b}{c}=\frac{\sin A+\sin B}{\sin C}\Rightarrow \sin C=1$

$\Rightarrow C=90^o$

$A+B=90^o\Rightarrow \cos A=\sin B$

43. C

Sol. $f(x)=1+x\int_0^1 e^y f(y)\,dy+e^x\int_0^1 y\,f(y)\,dy$

Let $A=\int_0^1 e^y f(y)\,dy,\ B=\int_0^1 y\,f(y)\,dy$

$\Rightarrow f(x)=1+Ax+Be^x\Rightarrow A=\int_0^1 e^x\left(1+Ay+Be^y\right)dy$

$B=-\frac{2}{e+1},A=-\frac{3}{2}$

$f'(x)+3>0$

$\Rightarrow -\frac{3}{2}-\frac{2e^x}{(e+1)}+3>0\Rightarrow e^x<\frac{3(e+1)}{4}\Rightarrow\left[\frac{4}{3}e^x\right]=[e+1]=3$

44. A

Sol. $Ax_1+Be^{x_1}=-\frac{3x_1}{2}-2\Rightarrow Be^{x_1}=-2$

$f'(x_1)=A+Be^{x_1}=m_1$

$m_1=-\frac{3}{2}-2=-\frac{7}{2}$

$m_2=-\frac{3}{2}$

$\tan\theta=\frac{8}{25}$

45. A

Sol. $\lim_{x\to1}\frac{f(x)}{(x-1)^2}=1$

$\Rightarrow f(1)=0,\ f'(1)=0,\ f''(1)=2$

$\therefore\ f(x)=A(x-1)^4+B(x-1)^3+(x-1)^2$

$f'(x) = 4A(x-1)^3 + 3B(x-1)^2 + 2(x-1)$

$f'(0) = -6 \Rightarrow -4A + 3B - 2 = -6$ .....(i)

$f'(2) = 6 \Rightarrow 4A + 3B + 2 = 6$ .....(ii)

(i), (ii) $\Rightarrow A = 1, B = 0$

$\therefore\ f(x) = (x-1)^4 + (x-1)^2 \geq 0$

$f(1) = 0$ is the minimum.

46. C

Sol. $f(x) = (x-1)^4 + (x-1)^2$, $f'(0) = -6$, $f(0) = 2$

$f'(x) = 4(x-1)^3 + 2(x-1)$

Subtangent at $x = 0$ is $\left|\frac{f(0)}{f'(0)}\right| = \frac{2}{6} = \frac{1}{3}$

SECTION – D

47. 00021.61

Sol. The number of ways when no student failed in any examination $= (2^3 - 1)^4$

The number of ways when out of above cases atleast one subject was not cleared by any students $= 3C_1(3)^4$

The number of ways when out of above cases atleast any two subjects were not cleared by any student $= 3C_2(1)^4$. So required cases $= (2^3-1)^4 - 3C_1 3^4 + 3C_2 = 2161$

48. 00006.28

Sol. $S = \alpha + \alpha^2 + \alpha^4 + \alpha^3 + \alpha^5 + \alpha^6 = -1$

$P = (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6)$

$= \alpha^4 + \alpha^5 + \alpha^7 + \alpha^6 + \alpha^7 + \alpha^9 + \alpha^7 + \alpha^8 + \alpha^{10}$

$= 3 + \alpha^4 + \alpha^5 + \alpha^6 + \alpha^2 + \alpha^1 + \alpha^3$

$= 3 - 1 = 2$

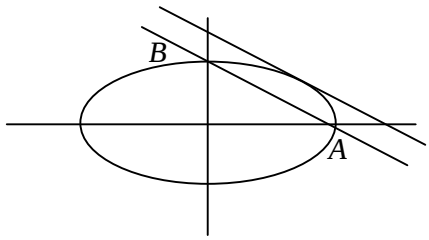
$\therefore\ c = 2$

49. 00003.00

Sol. Let perpendicular distance of *P* from the line be *h*

$\frac{1}{2} \times h \times 5 = 6(\sqrt{2} - 1)$ (as $\Delta PAB = 6(\sqrt{2}-1)$)

$\Rightarrow \quad h = \frac{12(\sqrt{2}-1)}{5}$



Now distance of tangent parallel to *AB* i.e.

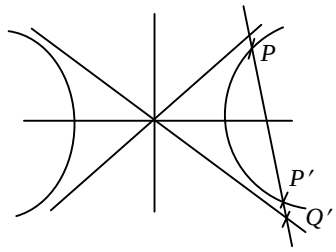
$4y + 3x = 12\sqrt{2}$, from line *AB* is $\frac{12(\sqrt{2}-1)}{5}$.

There are just three such points.

50. 00000.02

Sol. For any line in case of hyperbola we know that $PQ = P'Q'$

$$\frac{PQ}{P'Q'} + \frac{PQ'}{P'Q} = 1 + \frac{PP' + P'Q'}{PP' + PQ} = 2$$



51. 00013.33

Sol. Let $\vec{r} = \lambda\vec{b} + \mu\vec{c}$ and $\vec{c} = \pm(xi + yj)$. Since $\vec{c}$ and $\vec{b}$ are perpendicular, we have

$4x + 3y = 0 \qquad \Rightarrow \quad \vec{c} = \pm x\left(i - \frac{4}{3}j\right)$

$\pm 1 =$ proj. of $\vec{r}$ on $\vec{b} = \frac{\vec{r}.\vec{b}}{|\vec{b}|} = \frac{(\lambda\vec{b} + \mu\vec{c}).\vec{b}}{|\vec{b}|} = \frac{\lambda\vec{b}.\vec{b}}{|\vec{b}|}$

$\left[\because \vec{b}.\vec{c} = 0\right]$

$= \lambda|\vec{b}| = 5\lambda$. Hence $\lambda = \pm\frac{1}{5}$

Also, $\pm 2 =$ proj. of $\vec{r}$ on $\vec{c} = \frac{\vec{r}.\vec{c}}{|\vec{c}|}$

$= \frac{(\lambda\vec{b} + \mu\vec{c}).\vec{c}}{|\vec{c}|} = \mu|\vec{c}| = \frac{5}{3}\mu x$

Thus, $\mu x = \pm\frac{6}{5}$. Therefore,

$\vec{r} = \frac{1}{5}(4i + 3j) + \frac{6}{5}\left(i - \frac{4}{3}j\right) = \pm(2i - j)$

$\vec{r} = \frac{1}{5}(4i + 3j) - \frac{6}{5}\left(i - \frac{4}{3}j\right) = \pm\left(-\frac{2}{3}i + \frac{11}{5}j\right)$

Thus there are four such vectors

$\sum_{i=1}^{4}|\vec{r}_i|^2 = 2|2i - j|^2 + 2\left|-\frac{2}{5}i + \frac{11}{5}j\right|^2 = 20$

52. 00001.00

Sol. As f is continuous so $f(0) = \lim_{x\to 0} f(x) = \lim_{n\to\infty} f(x_n)$ for any sequence x_n such that

$\lim_{n\to\infty} x_n = 0$. Thus $f(0) = \lim_{n\to\infty} f\left(\frac{1}{4n}\right) = \lim_{n\to\infty}\left((\sin e^n)e^{-n^2} + \frac{1}{1 + \frac{1}{n^2}}\right) = 0 + 1 = 1$

$\left(\lim_{n\to\infty}(\sin e^n)e^{-n^2} = 0 \text{ as } \left|(\sin e^n)e^{-n^2}\right| \le e^{-n^2} \text{ and } e^{-n^2} \to 0 \text{ as } n \to \infty\right)$

53. 00000.50

Sol. Put $\frac{1}{x}=t$, so that $I=\int_{1/e}^{e}|\log x|(-1)dt$

$=\int_{e}^{1/e}\left|\log\frac{1}{t}\right|(-1)dt$

$=\int_{1/e}^{e}|-\log t|dt$

$=\int_{1/e}^{1}(-\log t)dt+\int_{1}^{e}(\log t)dt$

$=(-t\log t+t]_{1/e}^{1}+(t+\log t-t]_{1}^{e}$

$=1-\frac{1}{e}-\frac{1}{e}+e-e+1$

$=2\left(1-\frac{1}{e}\right)$

54. 00020.83

Sol. E_1 : First bag is chosen, $P(E_1)=\frac{1}{2}$.

E_2 : Second bag is chosen, $P(E_2)=\frac{1}{2}$.

A : Drawn number is 4.

Now, $P(A)=P(E_1).P\left(\frac{A}{E_1}\right)+P(E_2).P\left(\frac{A}{E_2}\right)$

$=\frac{1}{2}.\frac{1}{4}+\frac{1}{2}.\frac{1}{6}=\frac{5}{24}$